

Dynamics of microbial protein synthesis in the rumen - A Review

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ABSTRACT

Ruminants establish a symbiotic relationship with rumen microorganisms by which the animal provides nutrients and optimum environmental conditions for the fermentation of feeds, and microorganisms degrade fiber and synthesize microbial protein as an energy and protein supply for animal. Ruminant proteins degradation is affected by pH and predominant species of microbial population. A strong positive correlation was observed between dry matter intake (DMI) and microbial growth. The more carbohydrate digestion the more microbial protein synthesis occur. Fermentation energy is inversely related to microbial growth. Essential oil has domestic anti- microbial activity. Nitrogen compound or crude protein content (CP) of many practical diets may be greater than the 11% required to support optimal microbial growth. Forage concentrate mixing preparation increase the efficiency of microbial protein synthesis than only forage preparation. The efficiency of microbial protein synthesis greater in forages containing saponin and tannins which reduce ruminal N degradability. As ruminant out flow rate increase microbial protein synthesis, it is supposed that the efficiency of microbial protein synthesis can be increased by about 20% if rumen outflow rate is increased from 0.02 to 0.08 /h. Microbial protein production will be a function of the availability of vitamins, microminerals and protein level in the diet of digestible organic matter. Microbial protein synthesis is dependent upon suitable N and carbohydrate sources. Even though trace minerals and vitamins are adequate for maximal microbial protein synthesis in many feeding conditions, inadequate trace minerals and vitamins, in some cases, could limit microbial protein synthesis.

INTRODUCTION

Microorganisms in the rumen degrade nutrients to produce volatile fatty acids and synthesize microbial protein as an energy and protein supply for the ruminants. Ruminants establish a symbiotic relationship with rumen microorganisms by which the animal provides nutrients and optimum environmental conditions for the fermentation of feeds, and microorganisms degrade fiber and synthesize microbial protein as an energy and protein supply for animal. Rumen microbial protein represents a major source of amino acids to the ruminant animal. Microbial protein can supply from 70% to 100% of amino acids to ruminant (AFRC, 1992). High microbial protein production can decrease the need for supplementing rumen undegradable protein (Blummel et al., 1999). The amino acids reaching the small intestine are

supplied by the microbial protein, the undegraded feed protein, amino acids and peptides from feed which escape degradation, and endogenous secretions. The microbes that are produced in the rumen, and then pass down the digestive tract, may supply 60 to 80 percent of the amino acids absorbed from the small intestine. The efficiency of microbial protein synthesis is a major factor affecting the overall amino acid requirement of ruminants and caecum fermenters, and is influenced by a number of factors including energy source, supply of nutrients (nitrogen, sulfur, branched chain fatty acids) and rumen environmental characteristics such as dilution rate, pH and microbial species present (Caton et al., 1993). An average efficiency of microbial synthesis of 17 grams of microbial protein per 100 grams of digestible organic matter was determined for many diets, although values were generally higher for sheep and forage based diets than for cattle and concentrate diets (Beharka and Nagaraja, 1998). A continuous-culture experiment showed about a 14% to 30% increase in the amount of microbial protein available for use. Microbial protein contributes about two third of the amino

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acids absorbed by ruminants (Pathak, 2008). Although it is characterized by a relatively high proportion of non protein nitrogen (25%), (AFRC, 1992) it has an invaluable role in the nutrition of ruminant animals. Daily microbial protein synthesis is different from the efficiency of microbial protein synthesis (Hoover and Stokes, 1991), which usually is defined as grams of microbial crude protein (MCP) / kilogram or 100 grams of organic matter (OM) digested in the rumen (Hoover and Stokes, 1991). The amino acid composition of microbial true protein is similar to that of protein in the main animal products, such as milk, lamb and beef (Orskov, 1992). Compare to oil seed meal and legume grains microbial protein contains a higher proportion of methionine and lysine (DLG, 1976).

Hoover and Stokes (1991) proposed that the rate of digestion of carbohydrates would have greater impact on the microbial protein synthesis. The microbial protein synthesis is reported to be low in animals fed high-concentrate diets because of reduced ruminal pH (NRC, 1996). The microbial protein synthesis is also low in low-quality forages because of slow carbohydrate degradation; in situ data showed that the ratio of degraded nitrogen to organic matter in the rumen greatly varied in the rumen in times after feeding. It seems that diets containing a mixture of forages and concentrates increase the efficiency of microbial protein synthesis because of an improved rumen environment for the growth of more diverse bacteria species. This discussion focused on production of microbial protein in rumen prone to several dynamics.

The pH value: The pH value may alter the microbial protein yield in the rumen. Low pH may be deleterious to rumen microbes, and especially sensitive are protozoa. A low pH value is also expected to reduce the digestibility of fibrous plant tissues. Due to low pH, energy within the rumen is diverted to non growth functions i.e. maintaining neutral pH in bacterial cells

(Strobel and Russel, 1986). Ruminal protein degradation is affected by pH and the predominant species of microbial population. Ruminal proteolytic activity decreases as pH decreases with high-forage dairy cattle-type rations, but not in high-concentrate beef-type rations (Bach et al., 2005).

Rumen microflora: There are approximately 1-10 billion bacteria in each milliliter (ml) of gut or contents. On high-forage diets, there are normally between 1 to 3 billion bacteria per ml of contents, and on high-grain diets, there are normally between 8 to 10 billion bacteria per ml of contents. The main reason for the difference in concentration are that high-forage diets contain more lignin, which can limit the surface area of available polysaccharides. And because there is a 'lag time' for the bacteria to attach to the forage particles (Pathak, 2008). With high-grain diets, there is usually much more surface area for the bacteria to attach to, as grains contain very little lignin, and because the time it takes starch digesting bacteria to replicate is normally less than the time it takes cellulose digesting bacteria to replicate. Bacteria can replicate in 10 minutes to 2 hours, depending on the species. Any type of feed processing that increases the surface area available for bacterial attachment increases the number of bacteria digesting feed at any one time.

Wiedmeier et al. (1987) had shown that addition of stimulatory factor (readily fermentable carbohydrate) to rumen bacteria, resulting in a 27% increase in cellulolytic bacteria, with increased dry matter, and crude protein digestion as well as an increase in the acetate to propionate ratio. In addition, Frumholtz et al. (1989) resulted in a 179% increase in the total number of ruminal bacteria and a 288% greater number of cellulolytic bacteria with a 45% reduction in protozoal numbers compared with the control treatment lacking carbohydrate in one fermentation study. Concentrate addition to sheep diets has been shown to result in a doubling of

the number of total culturable bacteria (Fondevila et al., 1990). Beharka et al. (1991) stated that supplemented calves have been found to have higher bacterial counts of cellulolytic, hemicellulolytic, and pectinolytic bacteria. In other hand, Newbold et al. (1992) showed the result in 90% increase in total bacteria and 50% increase in cellulolytic bacteria, and was reported to stimulate both bacterial growth and activity. Campos-Montiel and Viniegra-Gonzalez (1995) found to result in a 12.6% increase in cellulase activity, with a 32.7% increase in acetic acid production showing an increase in both bacterial growth rate and metabolic activity. They also found supplementation increase in 17.4% cellulolytic bacteria protein biomass, showing an increase in both growth rate and metabolic activity. Meanwhile, Kreikemeier and Varel (1997) stated that supplementation increased cellulolytic bacteria in steers. Beharka and Nagaraja (1998) found increase growth rate of the fiber digesting bacteria in the rumen, *Fibrobacter succinogenes* S85 and *Ruminococcus albus* 7 as well as several strains of the lactate utilizing bacteria *Megasphaera elsdenii*, *Selenomonas ruminantium*, and *Selenomonas lactilytica*. Caton et al (1993) reported that supplementation forage increase intake and increase total, essential, and nonessential amino acid flows to the duodenum in steers grazing cool-season pasture, primarily smooth brome, from June through August.

Dry matter intake: A strong positive correlation was observed between dry matter intake (DMI) and microbial growth (Djouvinov and Todorov, 1994). Although increasing the level of intake decrease the percentage of organic matter digested in the rumen. Therefore, more nutrients were supplied for microbial growth. Increasing the DMI with the addition of straw to barley-based diets significantly increased microbial protein synthesis in the rumen (Pathak, 2008). Similarly, the supplementation of straw with starch linearly increased the amounts of OM

digested and solid and liquid out flow rates. Therefore, increasing the level of starch linearly increased microbial yields, resulting in a strong correlation between the digestible organic matter intake and the microbial protein synthesis. The increase in microbial protein synthesis with increased feed intake is probably the result of the increased passage rate (Pathak, 2008). The increased passage of microbial protein in the small intestine occurred as a result of the increased passage of both fluids and solids with increased intake (Djouvinov and Todorov, 1994). The higher level of dietary CP led to increase DM intake, rumen ammonia concentration, N retention, live weight gain and rate of urinary excretion of purine derivatives (Doan et al., 2009).

Organic matter of feed: A major energy source of organic matter is carbohydrate for microbial protein synthesis; some researchers have suggested that it would be more appropriate if the efficiency of microbial protein synthesis is expressed as a function of carbohydrate digested rather than organic matter digested in the rumen (Nocek and Russell, 1988). The efficiency of microbial protein synthesis greatly differs in animals fed different diets, even within similar diets. Pathak (2008) reported the average efficiency of microbial protein synthesis was 13.0 for forage based diets, 17.6 for forage concentrate mix diets, and 13.2 g MCP/100g for concentrate diets of OM truly digested in the rumen. Overall, the average efficiency of microbial protein synthesis was 14.8g MCP/100g of OM truly digested in the rumen.

Level of feeding: Experimental evidence is available which suggest that frequency of feeding improve the efficiency of microbial protein synthesis (Khandaker, 1998) and was certainly observed through stimulation models of rumen function (Baldwin and Denham, 1979). Tamminga (1979) reported an increase of about 20-30% protein flow in intestine with increasing feeding frequency of dairy cows, although sheep did not show the same result (MacRae et al., 1972).

Frequent feeding increases the rate of passage of liquid and solids from rumen (Sutton, 1980) and influence in microbial protein synthesis. Similarly feeding sheep at 2 hrs interval resulted in greater protein synthesis than feeding once daily (Al-Atter et al., 1976). Sutton (1980) reported that feeding frequency decrease the digestibility. However, many authors observed a positive effect of frequent feeding on microbial biomass production (McAllan and Smith, 1983, Djouvinov and Todorov, 1994).

Fermentable energy: Fermentable energy supply is usually the first limiting factor for microbial growth in the rumen. Microbial yield in rumen depends largely on the availability of carbohydrate and nitrogen in rumen (Chumpawadee et al., 2006). Nocek and Russel (1988) suggested that the efficiency of microbial growth and microbial protein production may be improved by balancing the overall daily ratio of ruminally available energy and N in the diet. Shabi et al. (1998) found that the available energy in the rumen (Ruminal degradable organic matter) is the most limiting factor for ruminal N utilization. To estimate the microbial protein yield, modern European protein systems use information, which is directly or indirectly used in estimating the energy supply to the animal. The microbial protein yield can be estimated on the basis of ME, NE, fermentable metabolizable energy, digestible carbohydrates or fermentable organic matter (Verbic and Babnik, 1997; GFE, 2001). Supplementary energy source up to 5% of total DMI, does not affect DMI, rumen ammonia concentration and nitrogen retention (Doan et al., 2009) but increase microbial protein synthesis and growth. They also stated that maize appears to support more efficiently synthesis of microbial protein than molasses due to more excretion rate of purine derivatives and projected higher live weight gain.

Firkins (1996) stated that microbial protein supply to the duodenum should be maximized for efficient use of feed protein

and energy. Increasing starch in the diet decreases ruminal pH, which often decreases extent of ruminal fiber digestion and also may decrease efficiency of microbial protein synthesis because of energy-spilling reactions. The maximum potential of rumen microbes to produce microbial protein can be explored only by the provision of high quality forage. The problem of low microbial protein yield in diets containing low quality forages can not simply be solved by supplementing diets with high amount of concentrates. It has been shown that in diets containing high level of concentrates the efficiency of microbial protein synthesis in the rumen is lower than in well-balanced forage based diets (ARC, 1984). The primary function of the microbial carbohydrate metabolism is to release the ATP required for microbial growth. Thus, patterns and rate of microbial N metabolism are dependent upon the rates of carbohydrate fermentation (Hoover and Stokes, 1991).

Essential oil: Borchers (1965) was the first to report the potential benefit of essential oils on rumen microbial fermentation and observed that the addition of thymol to rumen fluid in vitro resulted in the accumulation amino acid N and the reduction of ammonia concentration, suggesting that thymol inhibited deamination. However, since the announcement of the ban on antibiotics as feed additives in European Union (2006) there has been renewed interest in studying the effects and mechanism of action of essential oils on rumen microbial fermentation. Vakili et al. (2013) carried out an experiment on growing Holstein calf consuming high concentrate diet supplemented with essential oils and recommended that essential oils supplementation might be useful as ruminal fermentation modifiers with increased molar proportion of propionate in beef production system.

Nitrogen compounds: It is generally assumed that the amount of urea that should be added to the diet to optimize ruminal microbial growth, is equivalent to net microbial protein synthesis minus the degradable intake protein content of the basal diet divided by 2.8 (Burroughs et al., 1975). Microbial N flow to the small intestine is not enhanced when ruminal degradable N exceeds 75% of net microbial synthesis (Zinn and Shen, 1998). More importantly, there is considerable evidence (Zinn et al., 2003) that growth-performance of feedlot cattle may be enhanced by levels of urea supplementation in excess of that required to optimize microbial protein synthesis. The basis for this effect may be due to the alkalizing effects of urea as it is hydrolyzed within the rumen to form ammonium carbonate.

Nitrogen compound or crude protein contain (CP) of many practical diets may be greater than the 11% required to support optimal microbial growth; the resistance of proteins to microbial degradation may limit microbial protein synthesis (Pathak, 2008). Protein degradation in the rumen is one of the main regions for the inefficient utilization of protein in ruminants. On the other hand, the nitrogen compounds, which are released during the protein degradation, are crucial for microbial growth in the rumen. It seems that, protein which has lower rates of ruminal degradation tend to improve the efficiency of microbial protein synthesis, probably because of the better capture of released N by rumen microbes. In modern protein synthesis it is required that the need of rumen microbes for nitrogen compounds are fully covered either by degradable dietary protein or by metabolic nitrogen, which arise from the oxidation of amino acids in animal tissues and which can be recycled into the rumen (Pathak, 2008). In some system it is proposed that the capture of rumen degradable protein is not complete (AFRC, 1992) and therefore a surplus of rumen degradable protein is required. Dietary

forage have higher microbial protein yield, metabolizable protein and increased N conversion (Zhu et al., 2013). Under normal feeding condition, a large proportion of dietary protein will pass through the ammonia pool before it is utilized for microbial protein synthesis in the rumen (Hristov et al., 2005).

Efficiency of microbial protein synthesis (EMPS) is unable to estimate the efficiency at which bacteria capture available N in the rumen and determination of grams of microbial N per unit of rumen available energy (Bach et al., 2005). An alternative and complementary measure of microbial protein synthesis is the efficiency of N use (ENU). In contrast to EMPS, ENU is a good measurement for describing efficiency of N capture by ruminal microbes. Bach et al., (2005) concluded that optimum bacterial growth in the rumen occurs when EMPS is 29 g of bacterial N/kg of fermented organic matter, and ENU is 69%, implying that bacteria would require about 1.31 x rumen-available N per unit of bacterial N. Because the distribution of N within bacterial cells changes with rate of fermentation, AA N, rather than total bacterial N should be used to express microbial protein synthesis.

Balancing carbohydrates and proteins for optimum rumen microbial yield and degradation: Microbial growth depends on the amount and availability of nitrogen and energy. Ammonia utilization in the rumen is intrinsically related to carbohydrate availability (Russell et al., 1983). Ruminal proteolysis and deamination will proceed to complete conversion of dietary protein to ammonia if carbohydrate availability is low (Agle et al., 2010). Availability of carbohydrate increase, the ammonia production decreases as direct incorporation of amino N into microbial protein, thus bypassing the ammonia pool (Russell et al., 1983).

The major nutrients required by rumen microbes are carbohydrates and proteins, but the most suitable sources and quantities needed to support maximum growth have

not been determined (Hoover and Stokes, 1991). Based on the data from both the *in vitro* and *in vivo* studies, there is general agreement that rate of digestion of carbohydrates is the major factor controlling the energy available for microbial growth; in addition, rate of digestion of total carbohydrate is directly related to the proportion of starches, pectins, and sugars. Proteins affect both total fermentation and production of microbial DM per unit of carbohydrate fermented. It appears that the quantity of ruminally available protein needed to optimize microbial growth may, under some conditions, be as high as 14 to 15% of diet DM (Hoover and Stokes, 1991). Several *in vitro* (Henning et al., 1991) and *in vivo* (Cameron et al., 1991) studies demonstrated that infusions of increasing amounts of readily fermentable carbohydrate decreased ammonia N concentrations because of the improved N uptake by ruminal microbes. However, the optimum ratio of readily fermentable carbohydrate to ammonia N has not yet been determined. Hoover and Stokes (1991) suggested that in pH-controlled continuous culture fermenters, maximum microbial growth is attained with a 2:1 NFC: RDP ratio. Sahoo et al., (2010) reported that added starch reduced ammonia nitrogen in *in vitro* incubation trial.

UDP and RDP in diet: Khandaker et al. (2012) recommended that supplementation of RDP enhance the intake, digestibility and microbial protein synthesis which ultimately increases utilization of low-quality feed resources. Agle et al. (2010) concluded that high concentrate diet results in numerically greater utilization of ruminal ammonia N for microbial protein synthesis. Al-Dehnel et al. (1986) also reported greater incorporation of N recycled to the rumen into bacterial protein with a high concentrate (1:2, forage:concentrate ratio) versus a low concentrate (2:1, forage:concentrate ratio). The high concentrate diet results in considerably greater proportion of bacterial N in

duodenal digesta N implying greater microbial protein synthesis in the rumen (Agle et al., 2010). Hristov et al. (2005) carried out on dietary crude protein level and degradability on ruminal fermentation and nitrogen utilization and stated that excess RDP in diet could not be efficiently utilized for microbial protein synthesis and largely lost through urinary N excretion. Increasing CP content of animal diet may result not only in production (Wu and Satter, 2000), but also increase concentrations of ruminal ammonia and blood urea N and consequently occur greater urinary N losses (Castillo et al., 2001). Exemplified by Tamminga (1992) that ruminal N loss is the greatest single contributor to urinary N losses, metabolic losses, indigestible microbial N, losses in maintenance and inefficient conversion of absorbed amino acids milk/body protein may comprise up to 72% of the urinary N losses. If not used for microbial protein supply, RDP convert to ammonia, absorb through the ruminal wall, detoxified to urea in the liver (Lobley et al., 1995) and largely lost in urine; some RDP bypass the rumen and contribute to the duodenal AA and peptide flow (Choi et al., 2002).

Total digestible nutrients in feed: Hersom and Carter (2007) worked out the variation in CP, RDP, RUP, TDN, and RDP:TDN ratios among common co-product feedstuffs; the suggested optimal ratio of RDP:TDN is approximately 8–13% Hersom (2013). Generally, providing supplements with RDP:TDN ratios greater than this level ensures that an adequate level of RDP is available to rumen microorganisms, thus enhancing the utilization of low-quality forage by the animal. Generally, dry pregnant cows and even lactating cows with low levels of milk production can subsist on fairly low protein diets, i.e., less than 10% CP. However, as the level of production increases and cows lactate more heavily, their MP requirements increase, which may require an RUP increase, as well. The increase in RUP

requirement can also occur in young, fast-growing calves whose total protein requirements at times can be more than twice that of their dams. Huhtanen and Hristov (2009) demonstrated that ruminal protein degradability has no effect on milk protein yield and energy remains the most important dietary factor determining the intensity and efficiency of microbial protein synthesis in the rumen (Agle et al., 2010). Net synthesis of microbial N is not enhanced when ruminal degradable feed N is greater than 11% of dietary TDN (Zinn and Shen, 1998) or 8% of dietary DM (Zinn and Shen, 1998). Rumen microbes require a source of fermentable nitrogen, usually as ammonia although some species require preformed amino acids and peptides. The low nitrogen content of most mature grasses points to a need to combine them with a forage with a high N content. The efficiency of microbial protein synthesis was predicted to be around 13g MCP/100g of total digestible nutrient (TDN) for beef cows (Burroughs et al., 1974; NRC, 1996).

NRC (1985) summarized 16 experiments for cattle consuming >40% of their diet as forage to develop a microbial protein production equation ($MPg/d = 163.25 \times TDN \text{ kg/d} - 199.12$, $r^2=0.77$), the negative intercept implies that a negative microbial production is possible at low energy intake. However, the calculated the average microbial yield 106.7g of microbial true protein per kg of TDN. Burroughs et al. (1974) assumed that 104.4g of microbial true protein was produced per kg of TDN consumed. Rohr et al. (1986) summarized 22 experiments in which forage : concentrate ratio varied between 100:0 to 49:51 for the purpose of microbial protein production determination and stated the similar to that of Burroughs et al. (1974) production 104.4g of microbial true protein per kg of TDN intake. Microbial protein is related to ruminal digestible OM. The results of Hoover and Stokes (1991) suggest that degradable intake protein and nonstructural carbohydrates should be

considered rather than TDN alone when predicting microbial protein production.

Forage-concentrate ratio: Forage-concentrate ratio of diet is the vital factor to make the rumen microbes more efficient. The average efficiency of microbial protein synthesis was higher in forage-concentrate mix diets than for all forage diets (Pathak, 2008). Synthesis of microbial protein is improved by varying the source and degradability of energy incorporated into the diet (Sinclair et al., 1995). In contrast to results of Salter et al. (1983) several studies have reported increased utilization of ruminal ammonia nitrogen for microbial protein synthesis when diets contained readily digestible carbohydrates rather than starch in high fiber diets. As proposed by Hoover and Stokes (1991), the rate of carbohydrate digestion in diets and the synchronization of this rate with that of nitrogen release has an impact on microbial protein synthesis. Firkins (1996) found higher grain feeding increased efficiency of microbial protein synthesis because ruminal passage rate was increased. Microbial N synthesis was highest when highly ruminally available nonstructural carbohydrates were combined with highly ruminally available nonstructural carbohydrate were combined with poorly ruminally available protein. This situation would suggest that N utilization for forages having high readily degradable protein (RDP) will improve microbial growth when forages are supplemented with ruminally available nonstructural carbohydrates (Huber and Kung, 1981). Wieghart et al. (1986), however, reported no effect of a large increase in concentrate: forage ratio (from 35:65 to 80:20) on glucose fluxes across the gut and liver. Moorby et al. (2006) reported that a linear increase in total VFA and butyrate concentrate with increasing proportion of concentrate in dietary DM, but concentration of propionate was not affected.

Agle et al. (2010) found high concentrate decrease pH and ammonia concentration

and increase propionate. Acetate: propionate ratio was greater for low concentrate while DM intake, rumen methane production and microbial protein synthesis were unaffected by diet. NDF digestibility decreased in high concentrate. Oshio et al. (1987) and Sutton et al. (2003) reported that propionate concentration in ruminal fluid increased with the higher rate of grain fed animals and greater propionate production trigger an increase in gluconeogenesis. A high concentration of glucose in the rumen has been shown to decrease fiber digesting bacteria via the production of inhibitory compounds. Fibrolytic species may be inhibited by high concentrate diets, the rest of ruminal microflora increases in numbers (Oshio et al., 1987). When the diet is deficient in ruminally fermentable substrate, carbohydrate supplementation enhances microbial protein synthesis and production, irrespective of the type of carbohydrate (Hristove et al., 2005; Broderick et al., 2008). Urinary purine derivatives (PD) excretion increase by increasing the proportion of dietary concentrate (Moorby et al., 2006) suggesting enhanced microbial protein synthesis and outflow from the rumen.

Czerkawski, (1976) reported that sheep fed a diet composed of a mixture of hay and concentrate had greater microbial growth in the rumen compared to those fed concentrate and hay separately. The increase in microbial growth may have resulted from a better non-protein nitrogen to protein ratio in the mixed diet because the concentration of NPN is generally higher in forages than in concentrates. While forage may supply N as highly degradable protein or non protein N, concentrates may slowly supply N mainly as peptides and/or amino acids needed for microbial protein synthesis (Baldwin and Denham, 1979). It could also be caused by better utilization of amino acids and peptides in the mixed diet. Efficiency tends to be increased when readily fermentable carbohydrate is supplemented at less than 30% of the total

diet, but decreased when the supplementation level is greater than 70% (Huber and Kung, 1981). The decrease in efficiency of microbial protein passage to the small intestine when diets containing more than 70% concentrate are fed may occur because of a rapid rate of nonstructural carbohydrate degradation, resulting in an uncoupled fermentation (Polan, 1988). As the proportion of forage increases in dietary DM, there is greater saliva flow, a higher ruminal pH, improved cation exchange capacity, improved hydration, improved mat formation, leading to decreased retention times and greater microbial growth as microbial generation times are reduced (Sniffen and Robinson, 1987). Fresh grass optimizes the rumen environment. Efficient utilization of roughages depends on the availability of nutrients needed by both rumen microbes and the animal with maximizing feed intake and performance.

Fermentation rates of soluble sugars and starch provide higher level of ATP than structural carbohydrate up to 4 h post feeding, but they provide almost no ATP for microbial growth after 4 h post feeding. Approximately 3-4 h post feeding, cellulose and hemicelluloses degradation start and continue for long period (up to 96 h) post feeding, providing ATP for later microbial growth. Therefore feeding a mixture of forage and concentrate resulted in greater microbial protein synthesis compared to feeding only concentrate or forage (Pathak, 2008).

Forage containing saponin and tannins:

The efficiency of microbial protein synthesis greater in forages containing saponin and tannins which reduce ruminal N degradability. Saponins are found in different parts of the plants such as the roots, tuber, bark, leaves, seed and fruit. Saponins can kill or damage protozoa through their binding with sterols present in protozoal surface (Francis et al., 2002). Some researchers (Wang et al., 2000; Wina et al., 2005) reported that saponins reduced

anaerobic fungi in ruminant but some other researchers (Muetzel et al., 2003) did not find significant change in fungal concentration by using saponins from *S. pachycapra*. There is little information available on the effects of saponins on rumen bacteria. Wang et al. (2000) and Muetzel et al. (2003) concluded that saponins had a negative effect on Gram-positive bacteria but no effect on Gram-negative bacteria. Saponins can reduce ammonia concentration in rumen. Wina et al. (2005) reviewed 51 papers and observed 14 papers showed no effect and 17 papers showed negative effects on rumen ammonia. Tree leaves and some agro-industrial by-products used in livestock feed contain large amount of tannins. High level of tannins in diet (60-120 g/kg DM) may depress digestive efficiency and animal productivity (Patra, 2007). Tannins can reduce the populations of fibre-degrading *Ruminococcus* spp and *Fibrobacter* spp (McSweeney et al., 1999). Min et al. (2003) described that condensed tannins can reduce proteolysis and nitrogen loss during mastication and rumination. Min et al. (2003) reported that 20 to 45 g condensed tannin /kg dietary concentrate improved efficiency of N use and increased daily weight gain in lambs on temperate forage.

Lipid in diet: In rumen fermentation process has energy (loss of methane) and protein (loss of ammonia N) inefficiencies that may limit production performance and contribute to release of pollutants to the environment. Lipid of diet may contribute to reduce methane and ammonia loss from rumen. Lipid supplementation in the diet has become a promising strategy to increase the efficiency in animal production by an increase in the diet's energetic density (Hess et al., 2008). However, the incorporation of lipids in diets is limited since they are modified by rumen microorganisms that transform them through two important processes, namely lipolysis and biohydrogenation. The processes' intensity largely depends on supplement sources and

rates (Moate et al., 2004). Unsaturated fatty acids rates in the rumen above the saturation capacity of microorganisms produce adverse effects on rumen fermentation, such as decrease in fiber degradability, lowering of protozoa concentration, reduction in quantity and proportions of the short chain fatty acids (Silva et al., 2007). The above effects may be due to the fiber's physical lining with lipids (Jenkins and McGuire, 2006) and to the decrease in bacterial (in particular, gram-positive cellulolytic ones) and protozoa growth (Tamminga and Doreau, 1991). Decrease in the number of protozoa is generally associated with a reduction in bacterial N recycling in the rumen (Morais et al., 2006). Thus, an increase in the number of gram negative bacteria and a decrease in ammonia concentration occur (Jouany, 1996), with a possible increase in the flow rate of rumen solids (Czerkawski et al., 1975). These effects contribute towards an increase in the efficiency of microbial protein synthesis when lipids are supplemented for ruminants (Doreau and Ferlay, 1995).

Synchronized release of nitrogen and energy from diets: It has positive effect on microbial protein synthesis (MPS) reported by Dewhurst et al. (2001). They also concluded that the degree of synchrony in ruminal release of energy and nitrogen is likely to influence MPS only in diets containing adequate concentrations of readily fermentable carbohydrate. Zadeh et al. (2013) found that synchronization of rumen available protein and energy is one of the conceptual method to increase the efficiency of utilization of nutrients by the ruminants and maximum microbial protein synthesis could be achieved by matching the rate of organic matter and protein degradation. Firkins (1996) observed that ruminal degradation of carbohydrates and protein must be synchronized for optimal microbial efficiency, but the microbes appear to withstand transient periods of asynchronous nutrient supply in many cases.

Chumpawadee et al. (2006) showed that synchronizing the rate of dietary energy and nitrogen release DM digestibility trend increased, OM and ADF digestibility increased linearly. Finally they stated that synchronizing the rate of degradation of dietary energy and nitrogen release improves ruminal fermentation, microbial protein synthesis and feed utilization. Bacteria yield was increased by a synchronous supply of N and carbohydrates (Henning et al., 1991). Synchronization of the rate of dietary energy and nitrogen release has been suggested as a means of improving both microbial protein flow at the duodenum and the efficiency of microbial protein synthesis (Sinclair et al., 1995) a more efficient capture of rumen degraded nitrogen would reduce the excretion of urinary nitrogen (Sinclair et al., 1995) and fermentative carbon losses in CO₂ and CH₄ (Blummel et al., 1999).

Rumen out flow rate/ Rate of passage: It is one of the factors, which affect the efficiency of microbial protein synthesis in the rumen. Faster outflow rate is expected to reduce the maintenance costs of microbes because they spend less time within the rumen. In AFRC (1992) for instance, it is supposed that the efficiency of microbial protein synthesis can be increased by about 20% if rumen outflow rate is increased from 0.02 to 0.08 /h. Rumen outflow rate is a function of DM intake and therefore it can be assumed that the efficiency of microbial protein synthesis in the rumen can be increased in DM intake. One of the most important factors, which limits intake of low quality roughages, is their slow rate of degradation in the rumen. High quality roughages are therefore expected not only to increase microbial protein yield by providing high amounts of fermentable substrate but also by increasing the level of intake.

Vitamins and microminerals: Microbial protein production will be a function of the availability of vitamins, microminerals and protein level in the diet of digestible organic matter (ARC, 1984). In addition to N and

carbohydrate supply, microbial yield is affected by the concentrations of trace minerals especially dietary sulphur concentration and vitamins (Sniffen and Robinson, 1987). The amount of sulphur required by rumen microorganisms for synthesis of methionine and cysteine ranges from 0.11 to 0.20 % of the total diet, depending on the status of the cattle (NRC, 1996). Limited intake of sulphur may restrict microbial protein synthesis when large amount of NPN are fed to ruminant animals (Buttery, 1977) such as urea. Microbial biomass contain 8 g S/kg dry matter. An optimum nitrogen : sulphur ratio reported by Khandaker (1998) was 10:1 while McAllan and Smith (1983) reported 8.6–30.8:1. In addition, Bray and Till (1975) suggested 10-14:1 for sheep and for cattle 14-15:1. However, ARC (1980) recommended level is 14:1. Phosphorus is another mineral required for synthesis of ATP and nucleic acids (DNA, RNA) by rumen microbes. Mahfuz et al. (2014) reported higher nutrient digestibility in phosphorus group where 5g TSP was added with control feeding. Phosphorus supplementation acted as a mediator of higher digestion and metabolism of nutrient. Improvement in fibre degradation by phosphorus supplementation occurred through its specific stimulation on growth of rumen cellulolytic bacteria and anaerobic rumen fungi (Khan and Chaudhry, 2012). Since Phosphorus takes part in formation of DNA and RNA, the number of rumen microbes may increased at presence of TSP and ultimately the digestibility increased (Mahfuz et al., 2014).

Conclusion: N compound in ruminant diets serves as a source of metabolizable protein providing both ruminal-degraded protein for microbial protein synthesis and ruminal un-degradable protein. Microbial protein synthesis is dependent upon suitable N and carbohydrate sources. Even though trace minerals and vitamins are adequate for maximal microbial protein synthesis in many feeding conditions, inadequate trace

minerals and vitamins, in some cases, could limit microbial protein synthesis. Dietary protein sources, which are low in dietary protein intake, may limit the microbial protein synthesis when calculated to meet animal requirements based on dietary CP. In order to obtain maximal microbial protein synthesis, the N requirement of the rumen bacteria has to be met first in every respect. N sources also must include amino acids and peptides in addition to NPN. Frequency of feeding with diet containing a mixture of forages and concentrates increase microbial protein synthesis because of improved synchronization of nutrient release, an improved ruminal environment for more diverse ruminal bacteria species, increased amounts and type of substrates, increased intake and subsequently, increased rates of solid and liquid passage.

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